BASEL III Market Risk Modelling Using Monte Carlo VaR

Regulatory Framework
- Basel III
- BCBS Guidelines
- Internal Model Validation

Model Limitations
- Linear assumption
- Normality assumption
- Tail Risk underestimation

Factor-Based Monte Carlo VaR
CAPM-Style Model
XLK, MTUM, VLUE
Multivariate Normal

Return-Based Monte Carlo VaR
· Normal Distribution
· Mean & Volatility
· 95% 1-Day VaR

Probability Space:
Ω : Outcomes
E : Events
P : Probability Measure
X : Ω → R

Random Variables

VaR Methodologies

Uses mean & volatility

Normal Var:
$VaR_a = -(\mu + z_a \sigma)$

Definition of VaR
VaRα=inf{x:FL(x)≥α}

Confidence level:
α = 95%

F_t ~ N(μ_F , Σ_F)

Monte Carlo Simulation
- Random Sampling
- 10,000 Simulations
- Loss Distribution
- Tail Risk (5%)

F_1t : Tech Factor Return

F_2t : Momentum Factor Retur

F_3t :Value Factor Return

Systematic Risk factor returns

are treated as random variables to capture uncertainty in market

Portfolio Returns

are generated via Monte Carlo simulation to estimate the loss

Simulated (future) portfolio returns

ε : Ω → R

Idiosyncratic Risk

$R_t = \beta^T F_t + \varepsilon_t$

Expanded form

$R_t = \beta_1 F_{1,t} + \beta_2 F_{2,t} + \beta_3 F_{3,t} + \varepsilon_t$

$L_t = -R_t$

Portfolio Loss

Generates empirical loss distribution

CDF OF LOSS
$FL(x)=P(L \le x) = \Phi((x - \mu_l) / \sigma_l)$
$\Phi(z) = \int$ from $-\infty$ to z $[1 / \sqrt{(2\pi)}] e^{(-u^2 / 2)} du$

PDF:
$f_L(x) = 1 / sqrt(2\pi\sigma_L^2) \cdot exp(- (x - \mu_L)^2 / (2\sigma_L^2))$

Depends On

Defines Portfolio for

Portfolio Construction
· AAPL, MSFT, NVDA
· GOOG, AMD
· Fixed Weights
· Tech Equity Portfolio

computes return for

Return Calculation
· Adjusted Close Prices
· Log Returns
· Portfolio Returns

parametrized by

Statistical Estimation
Mean
Volatility
Covariance Matrix

Factor Moments:
μ_F=E[F_t],Σ_F=Cov(F_t)

Validates

Backtesting & Validation

Distributional Assumptions:
- Factor returns vector:
F_t ~ N(mu_F, Sigma_F)
- Mean of factors:
mu_F = E[F_t]
- Covariance matrix of factors:
Sigma_F = Cov(F_t) = E[(F_t - mu_F)(F_t - mu_F)^T]
- Distribution of Portfolio Return: R_t = β^T F_t + ε_t
R_t ~ N(β^T mu_F , β^T Sigma_F β + sigma_epsilon^2)
- Mean of portfolio return: mu_R = β^T mu_F
- Variance of portfolio return: sigma_R^2 = β^T Sigma_F β + sigma_epsilon^2

- Historical Backtesting
- Rolling Backtesting (250 Days)
- VaR Exceptions

Is Evaluated Using

Basel Traffic Light
- Green (0–4)
- Yellow (5–9)
- Red (10+)

Ensures Compliance With